13.12 | Moving Average Process MA(q)

Moving Average (MA) processes are regression models in which past residuals are used for forecasting future values of the time-series data. The Moving Average process is different from moving average technique discussed in Section 13.4, except that the regression model of the MA process can be considered as weighted moving average of past residuals. Moving Average process of lag 1, MA(1), is given by

$$Y_{t+1} = \mu + \alpha_1 \varepsilon_t + \varepsilon_{t+1} \tag{13.41}$$

Alternatively, a moving average process of lag 1 can be written as

$$Y_{t+1} - \mu = \alpha_1 \varepsilon_t + \varepsilon_{t+1} \tag{13.42}$$

MA(1) process uses the previous residual, ε_t, to forecast the next value of the time series. The reasoning behind MA process is that the error (also called shock or innovation) at the current period, ε_t, and the error at the next period, ε_{t+1}, drive the next value of the time series Y_{t+1}. A Moving Average process with q lags, MA(q) process, is given by

$$Y_{t+1} = \mu + \alpha_1 \varepsilon_t + \alpha_2 \varepsilon_{t-1} + \cdots + \alpha_q \varepsilon_{t-q+1} + \varepsilon_{t+1} \tag{13.43}$$

The value of q (number of lags) in a Moving Average process can be identified using the following rule (Yaffee and McGee, 2000):

1. Auto-correlation value, $|\rho_p| > 1.96/\sqrt{n}$ for first q values (first q lags) and cuts off to zero.
2. The partial auto-correlation function, ρ_{pk}, decreases exponentially.

13.13 | Auto-Regressive Moving Average (ARMA) Process

Auto-Regressive Moving Average (ARMA) is a combination of the Auto-Regressive and Moving Average process. ARMA(p, q) process combines AR(p) and MA(q) processes. ARMA(p, q) model is given by

$$Y_{t+1} = \overbrace{\beta_1 Y_t + \beta_2 Y_{t-1} + \ldots + \beta_p Y_{t-p+1}}^{\text{Auto Regressive Part}} + \overbrace{\alpha_1 \varepsilon_t + \alpha_2 \varepsilon_{t-1} + \ldots + \alpha_q \varepsilon_{t-q+1}}^{\text{Moving Average Part}} + \varepsilon_{t+1} \tag{13.44}$$

The parameters are estimated using Box–Jenkins methodology. The values of p and q in a ARMA process can be identified using the following thumb rule:

1. Auto-correlation value, $|\rho_p| > 1.96/\sqrt{n}$ for first q values (first q lags) and cuts off to zero.
2. Partial auto-correlation function, $|\rho_{pk}| > 1.96/\sqrt{n}$ for first p values and cuts off to zero.

Example 13.5

The monthly demand for avionic system spares used in Vimana 007 aircraft is provided in Table 13.24. Build an ARMA model based on the first 30 months of data and forecast the demand for spares for months 31 to 37. Comment on the accuracy of the forecast.

Table 13.24 | Demand for avionic spare parts

Month	Demand for Spares	Month	Demand for Spares
1	457	20	516
2	439	21	656
3	404	22	558
4	392	23	647

(Continued)

Table 13.24 | (*Continued*)

Month	Demand for Spares	Month	Demand for Spares
5	403	24	864
6	371	25	610
7	382	26	677
8	358	27	609
9	594	28	673
10	482	29	400
11	574	30	443
12	704	31	503
13	486	32	688
14	509	33	602
15	537	34	629
16	407	35	823
17	523	36	671
18	363	37	487
19	479		

Solution

The first step in building an ARMA model is model identification using ACF and PACF plots. The ACF and PACF plots based on the first 30 months demand data are given in Figures 13.9 and 13.10.

In Figure 13.9, the auto-correlations cuts off to zero after two lags (note that auto-correlation of lag 3 is just below the critical value). The PACF value cuts off to zero after the first lag. So, the appropriate model could be ARMA(1, 2) process. The SPSS output for ARMA(1, 2) process is given in Tables 13.25 and 13.26.

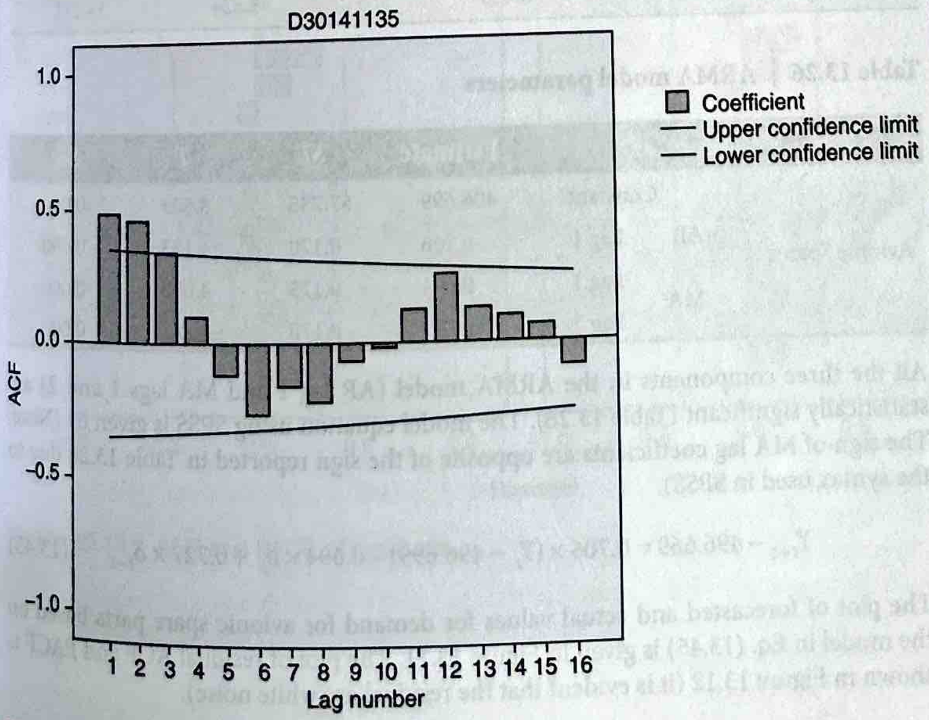


Figure 13.9 | ACF plot for avionic system spares demand.

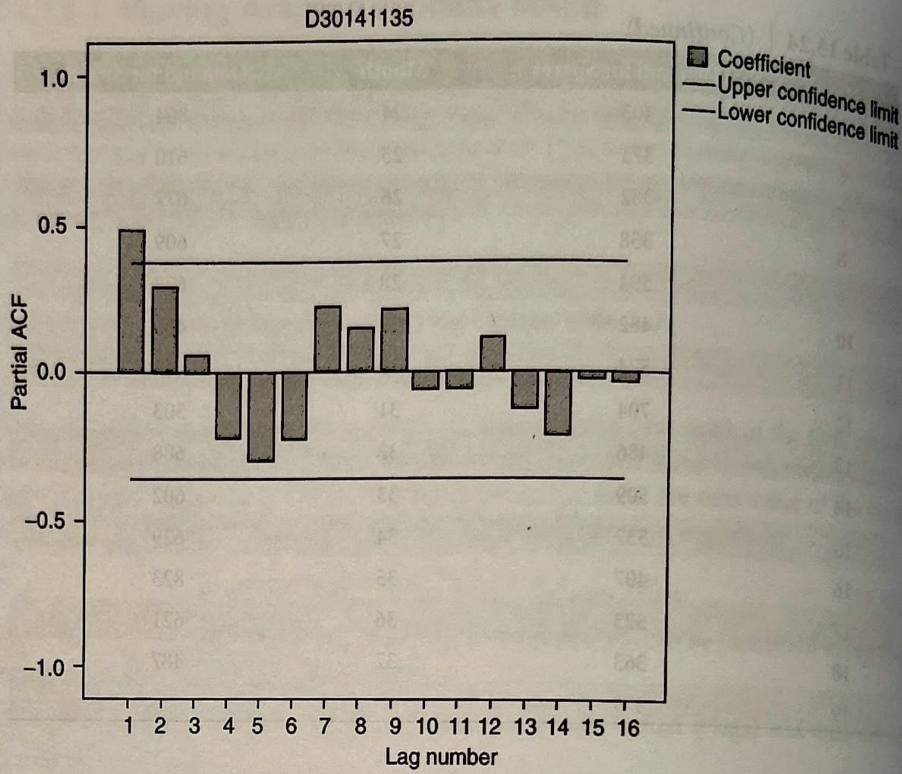


Figure 13.10 | PACF plot for avionic system spares demand.

Table 13.25 | Summary statistics

	Model Fit Statistics		
Model	Stationary *R*-Squared	RMSE	MAPE
Avionic Spares	0.429	98.824	14.231

Table 13.26 | ARMA model parameters

			Estimate	*SE*	*T*	Sig.
Avionic Spares		Constant	496.699	57.735	8.603	0.000
	AR	Lag 1	0.706	0.170	4.153	0.000
	MA	Lag 1	0.694	0.173	4.006	0.000
		Lag 2	−0.727	0.170	−4.281	0.000

All the three components in the ARMA model (AR lag 1 and MA lags 1 and 2) are statistically significant (Table 13.26). The model equation using SPSS is given by (Note: The sign of MA lag coefficients are opposite of the sign reported in Table 13.26 due to the syntax used in SPSS).

$$Y_{t+1} - 496.669 = 0.706 \times (Y_t - 496.699) - 0.694 \times \delta_t + 0.727 \times \delta_{t-1} \quad (13.45)$$

The plot of forecasted and actual values for demand for avionic spare parts based on the model in Eq. (13.45) is given in Figure 13.11. The plot of residual ACF and PACF is shown in Figure 13.12 (it is evident that the residual are white noise).

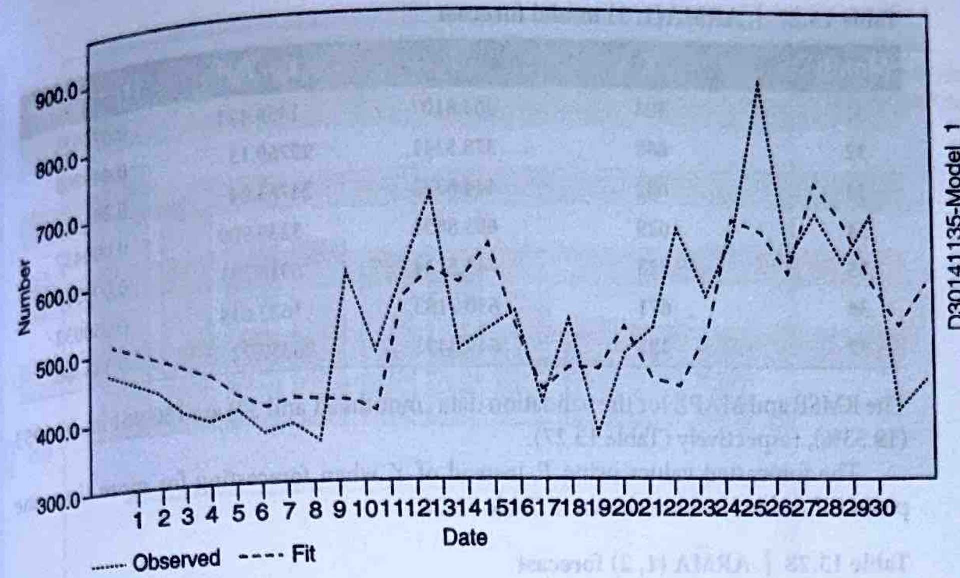


Figure 13.11 | Observed versus forecasted demand.

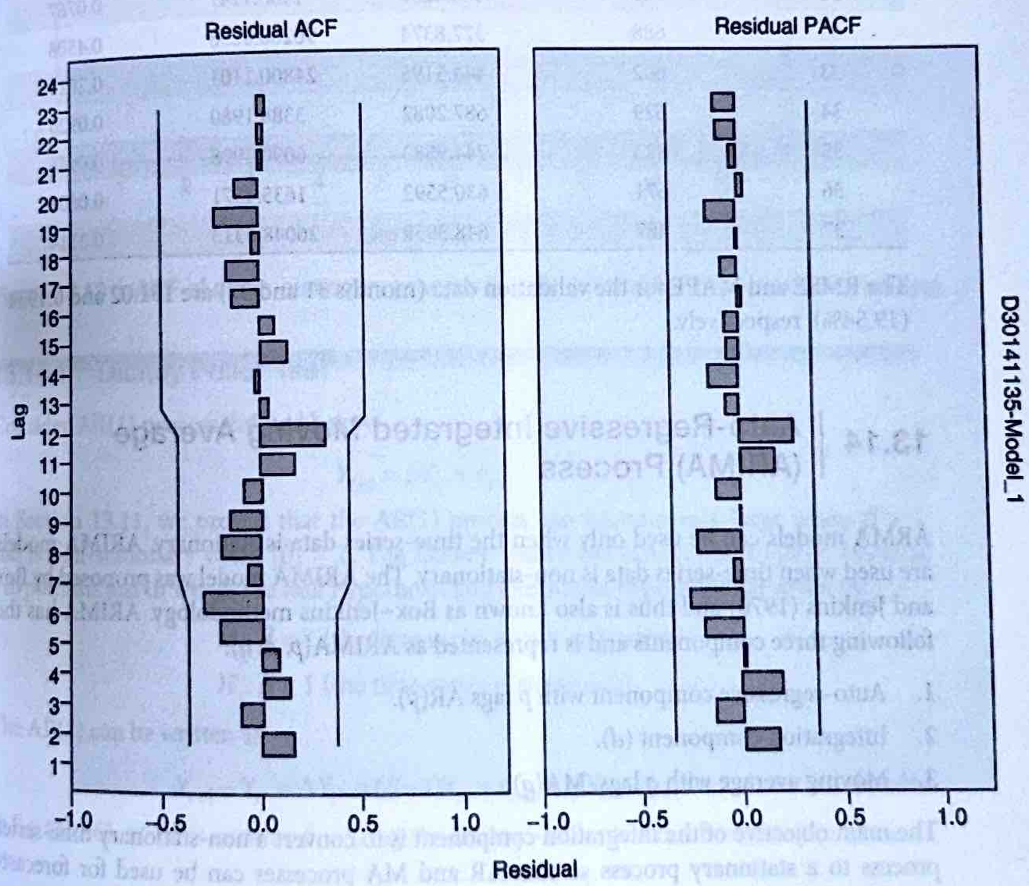


Figure 13.12 | ACF and PACF of residuals.

Table 13.27 | ARMA(1, 2) model forecast

Month	Y_t	F_t	$(Y_t - F_t)^2$	$\lvert Y_t - F_t \rvert / Y_t$
31	503	464.8107	1458.423	0.075923
32	688	378.5341	95769.15	0.449805
33	602	444.6372	24763.04	0.2614
34	629	685.8851	3235.909	0.090437
35	823	743.5124	6318.281	0.096583
36	671	630.7183	1622.614	0.060032
37	487	649.3491	26357.22	0.333366

The RMSE and MAPE for the validation data (months 31 and 37) are 150.961 and 0.1953 (19.53%), respectively (Table 13.27).

The forecasted values using F_t instead of Y_t when forecasting for more than one period ahead in time are shown in Table 13.28.

Table 13.28 | ARMA (1, 2) forecast

Month	Y_t	F_t	$(Y_t - F_t)^2$	$\lvert Y_t - F_t \rvert / Y_t$
31	503	464.4239	1488.1147	0.0767
32	688	377.8374	96200.8258	0.4508
33	602	444.5195	24800.1101	0.2616
34	629	687.2082	3388.1980	0.0925
35	823	744.9583	6090.4998	0.0948
36	671	630.5592	1635.4571	0.0603
37	487	648.3959	26048.6313	0.3314

The RMSE and MAPE for the validation data (months 31 and 37) are 151.02 and 0.1954 (19.54%), respectively.

13.14 | Auto-Regressive Integrated Moving Average (ARIMA) Process

ARMA models can be used only when the time-series data is stationary. ARIMA models are used when time-series data is non-stationary. The ARIMA model was proposed by Box and Jenkins (1970) and thus is also known as Box–Jenkins methodology. ARIMA has the following three components and is represented as ARIMA(p, d, q):

1. Auto-regressive component with p lags AR(p).
2. Integration component (d).
3. Moving average with q lags, MA(q).

The main objective of the integration component is to convert a non-stationary time-series process to a stationary process so that AR and MA processes can be used for forecasting. When data is non-stationary, the auto-correlation function will not be cut-off to zero quickly; rather ACF may show a very slow decline. When the time series is non-stationary, model parameter values will be greater than or equal to one resulting in non-convergence of the series and causing a slow decrease in the values of ACF and PACF. Thus, the presence of non-stationarity can be identified by plotting ACF. A slow decrease in ACF can be diagnosed as non-stationary process. Sample ACF plot for non-stationary data is shown in Figure 13.13.

Non-stationary could arise from deterministic or stochastic trends; identification of the source of non-stationarity will be useful for using appropriate transformation to make the process stationary. In addition to the visual evidence from ACF plot, Dickey–Fuller or augmented Dickey–Fuller tests are used to check the presence of stationarity.

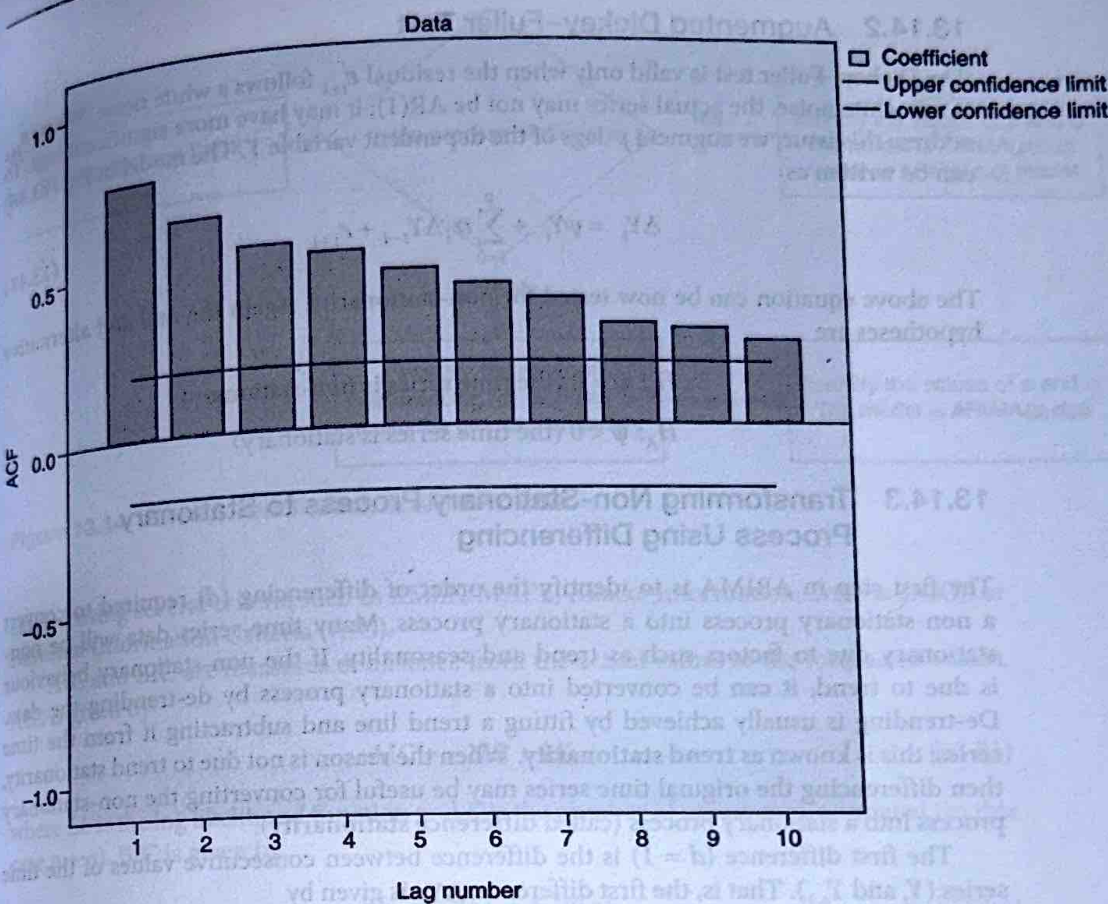


Figure 13.13 | ACF of a non-stationary process (slowly decreasing auto-correlation values).

13.14.1 Dickey Fuller Test

Consider AR(1) process defined below:

$$Y_{t+1} = \beta Y_t + \varepsilon_{t+1}$$

In Section 13.11, we proved that the AR(1) process can become very large when $\beta > 1$ and is non-stationary when $|\beta| = 1$. The Dickey–Fuller test (Dickey and Fuller, 1979) is a hypothesis test in which the null hypothesis and alternative hypothesis are given by

H_0: $\beta = 1$ (the time series is non-stationary)

H_1: $\beta < 1$ (the time series is stationary)

The AR(1) can be written as

$$Y_{t+1} - Y_t = \Delta Y_t = (\beta - 1)Y_t + \varepsilon_{t+1} = \psi Y_t + \varepsilon_{t+1} \tag{13.46}$$

In Eq. (13.46), $\psi = 0$ is same as $\beta = 1$. So, the Dickey–Fuller test can be written in terms of ψ as

H_0: $\psi = 0$ (the time series is non-stationary)

H_A: $\psi < 0$ (the time series is stationary)

The test statistic is given by

$$\text{DF Test Statistic} = \frac{\hat{\psi}}{S_e(\hat{\psi})} \tag{13.47}$$

where S_e is the standard error. Note that DF test statistic is not *t*-statistic since the null hypothesis is on non-stationary process. Critical values are derived based on simulation (Fuller, 1976).

13.14.2 Augmented Dickey–Fuller Test

The Dickey–Fuller test is valid only when the residual ε_{t+1} follows a white noise. When ε_{t+1} is not white noise, the actual series may not be AR(1); it may have more significant lags. To address this issue, we augment p-lags of the dependent variable Y. The model in Eq. (13.46) can be written as

$$\Delta Y_t = \psi Y_t + \sum_{i=0}^{p} \alpha_i \Delta Y_{t-i} + \varepsilon_{t+1} \tag{13.48}$$

The above equation can be now tested for non-stationarity. Again the null and alternative hypotheses are

H_0: $\psi = 0$ (the time series is non-stationary)

H_A: $\psi < 0$ (the time series is stationary)

13.14.3 Transforming Non-Stationary Process to Stationary Process Using Differencing

The first step in ARIMA is to identify the order of differencing (d) required to convert a non-stationary process into a stationary process. Many time-series data will be non-stationary due to factors such as trend and seasonality. If the non-stationary behaviour is due to trend, it can be converted into a stationary process by de-trending the data. De-trending is usually achieved by fitting a trend line and subtracting it from the time series; this is known as **trend stationarity**. When the reason is not due to trend stationarity, then differencing the original time series may be useful for converting the non-stationary process into a stationary process (called **difference stationarity**).

The first difference (d = 1) is the difference between consecutive values of the time series (Y_t and Y_{t-1}). That is, the first difference ΔY_t is given by

$$\Delta Y_t = Y_t - Y_{t-1} \tag{13.49}$$

The second difference (d = 2) is the difference of the first differences and is given by

$$\Delta^2 Y_t = \Delta(\Delta Y_t) = Y_t - 2Y_{t-1} + Y_{t-2} \tag{13.50}$$

In most cases, $d \le 2$ will be sufficient to convert a non-stationary process in to a stationary process.

13.14.4 ARIMA(*p*, *d*, *q*) Model Building

The first step in ARIMA(p, d, q) is the model identification, that is, identifying the values of p, d, and q. Box and Jenkins (1970) proposed the following procedure to build the ARIMA(p, d, q) model.

Stage 1: Model Identification

The main objective of the model identification stage is to identify the right values of p (auto-regressive lags), d (order of differencing), and q (moving average lags). The following flow chart can be used during the model identification stage (Figure 13.14). The first step is to plot the ACF and PACF to identify whether the time series is stationary or not. If the time series is stationary then d = 0 and the model is ARIMA(p, 0, q) or ARMA(p, q) model. If the time series is non-stationary then it has to be converted into a stationary process by identifying the order of differencing. Once the value of d is known, it will make the process stationary, then p and q are identified for the stationary process.

Stage 2: Parameter Estimation and Model Selection

Once the model is identified (values of p, d, and q), the next step in ARIMA model building is the parameter estimation. That is, the estimation of coefficients in AR and MA components which are achieved using ordinary least squares. Model selection may be

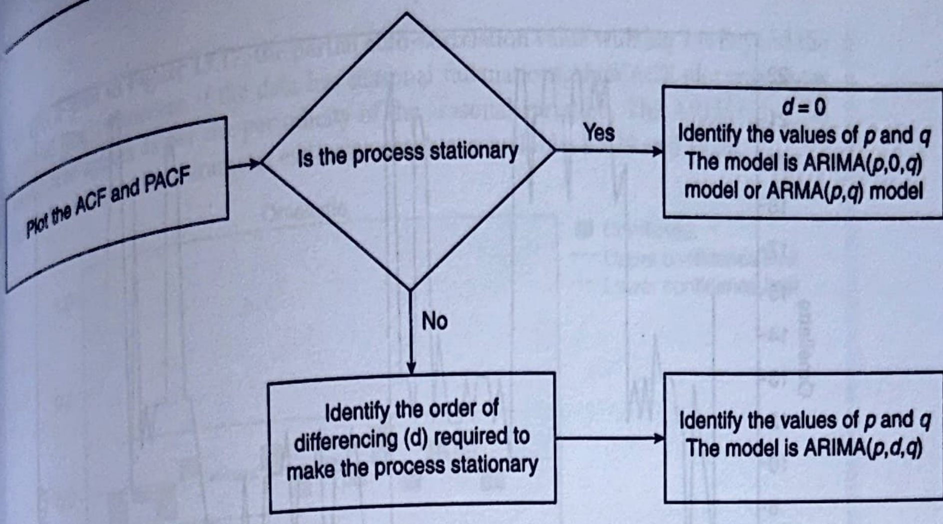


Figure 13.14 | Model identification in ARIMA model.

carried using several criteria such as RMSE, MAPE, Akaike Information Criteria (AIC), or Bayesian Information Criteria (AIC).

AIC and BIC are measures of distance from the actual values to the forecasted values. AIC is given by

$$AIC = -2LL + 2K \tag{13.51}$$

where LL is the log likelihood function and K is the number of parameters estimated (in this case $p + q$). BIC is given by

$$BIC = -2LL + K\ln(n) \tag{13.52}$$

In BIC equation, n is the number of observations in the sample. BIC assigns higher penalty compared to AIC for every additional variable added to the model. Lower values of AIC and BIC are preferred.

Stage 3: Model Validation

ARIMA model is a regression model and thus has to satisfy all the assumptions of regression. The residual should be white noise. We can also perform a goodness of fit test using Ljung–Box test before accepting the model.

Example 13.6

The daily demand for omelette at Die Another Day (DAD) hospital for the past 115 days is given in the excel sheet Example 13.6.xlsx. Develop an appropriate ARIMA model that DAD hospital can use for forecasting demand for omelette.

Solution

The time-series plot of the daily demand for omelette is shown in Figure 13.15. The corresponding ACF plot is shown in Figure 13.16. From Figure 13.15, it is evident that the mean is not constant for different values of t.

Since the ACF plot shows a very slowly decreasing pattern, we may conclude that the time series is not stationary. We have to convert the process to a stationary process before we can develop a forecasting model. The ACF and PACF plots after differencing ($d = 1$) are shown in Figures 13.17 and 13.18, respectively.

Since both ACF and PACF values are cutting off to zero after the first difference, we may conclude that the appropriate model is ARIMA(1,1,1). Note that subsequent correlations once it cuts off to zero are not useful and we will ignore them (for example,

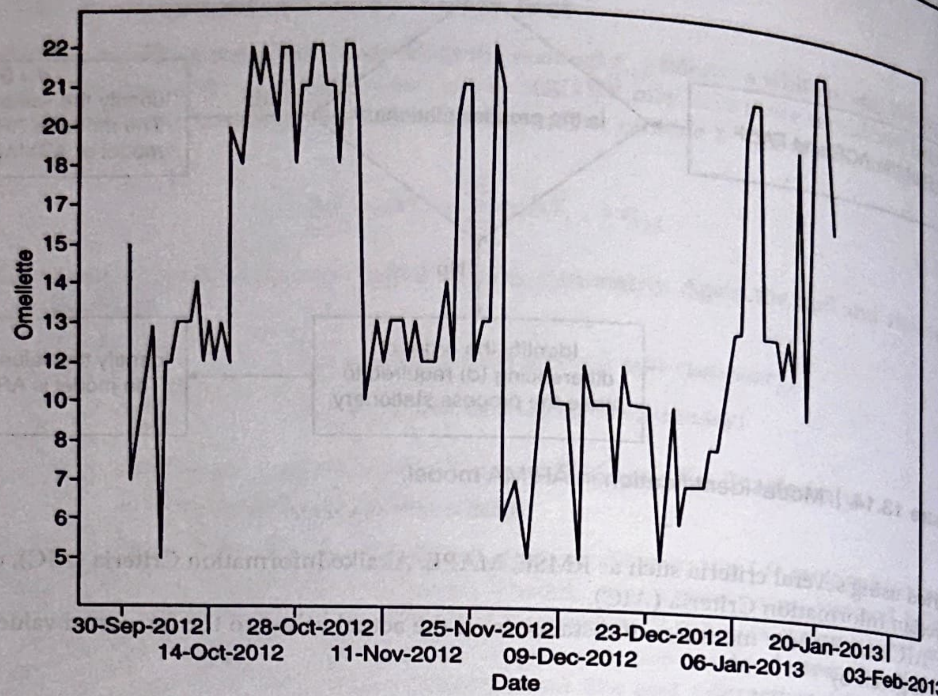


Figure 13.15 | Time-series plot of demand for Omelette at DAD hospital.

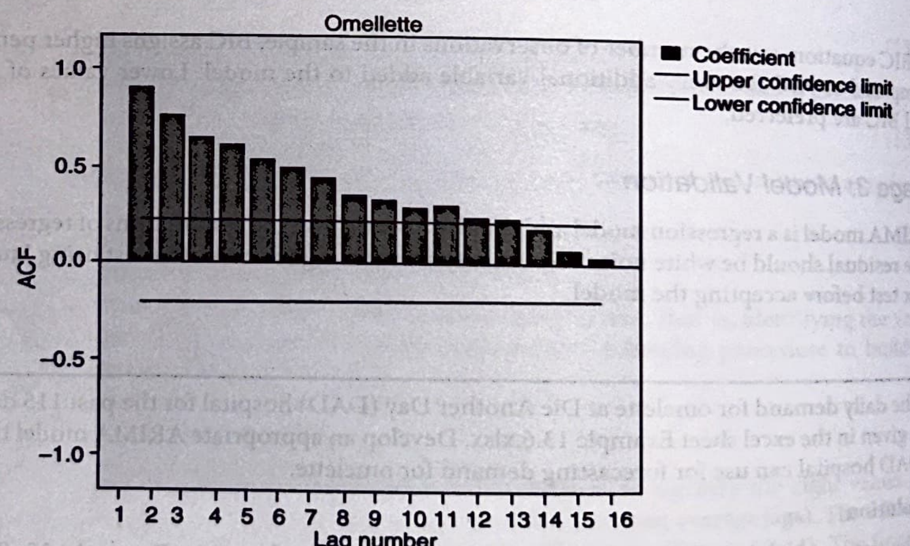


Figure 13.16 | ACF plot of demand for Omelette at DAD hospital.

in PACF plot in Figure 13.17, the partial auto-correlation value with lag 3 is beyond the critical line). However, if the data has seasonal fluctuations, then ACF plot may show consistent spikes as per the periodicity of the seasonal variation. The ARIMA(1, 1, 1) model summary and parameter estimates are shown in Tables 13.29 and 13.30.

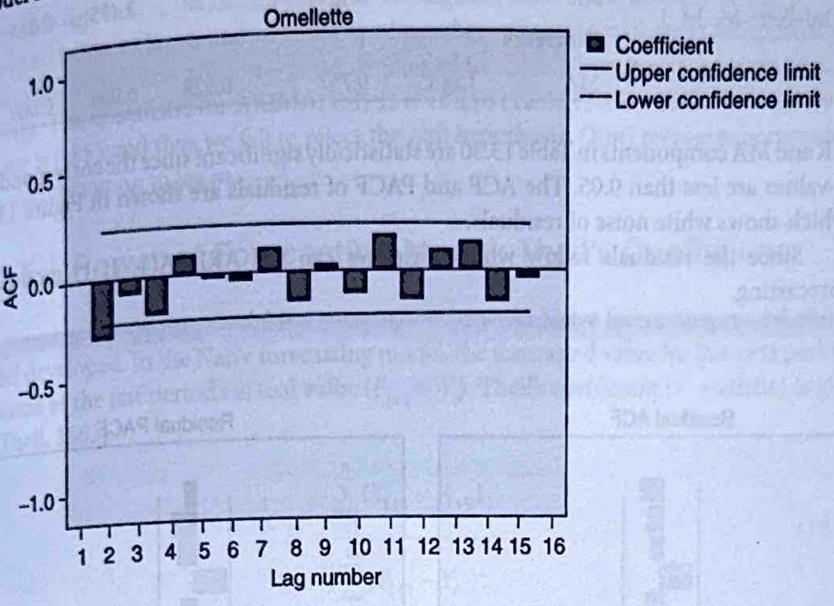


Figure 13.17 | ACF plot of demand for Omelette after differencing (d = 1).

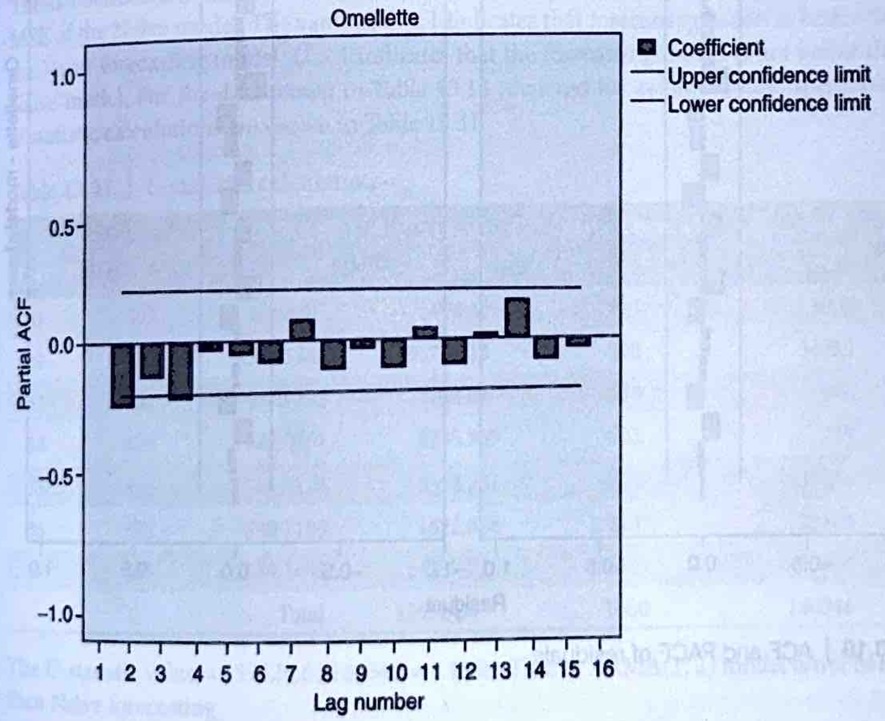


Figure 13.18 | PACF plot of demand for Omelette after differencing (d = 1).

Table 13.29 | ARIMA(1, 1, 1) model summary for Omelette demand

	Model Fit Statistics			Ljung–Box Q(18)		
Model	R-Squared	RMSE	MAPE	Statistics	Df	Sig.
Omellette-Model_1	0.584	3.439	20.830	10.216	16	0.855

Table 13.30 | ARIMA model parameters

			Estimate	SE	T	Sig.
Omellette-Model_1	Constant		0.055	0.137	0.402	0.689
	AR	Lag 1	0.439	0.178	2.475	0.015
	Difference		1			
	MA	Lag 1	0.767	0.128	6.004	0.000

AR and MA components in Table 13.30 are statistically significant since the corresponding p-values are less than 0.05. The ACF and PACF of residuals are shown in Figure 13.19 which shows white noise of residuals.

Since the residuals follow white noise, we can use ARIMA(1, 1, 1) model for forecasting.

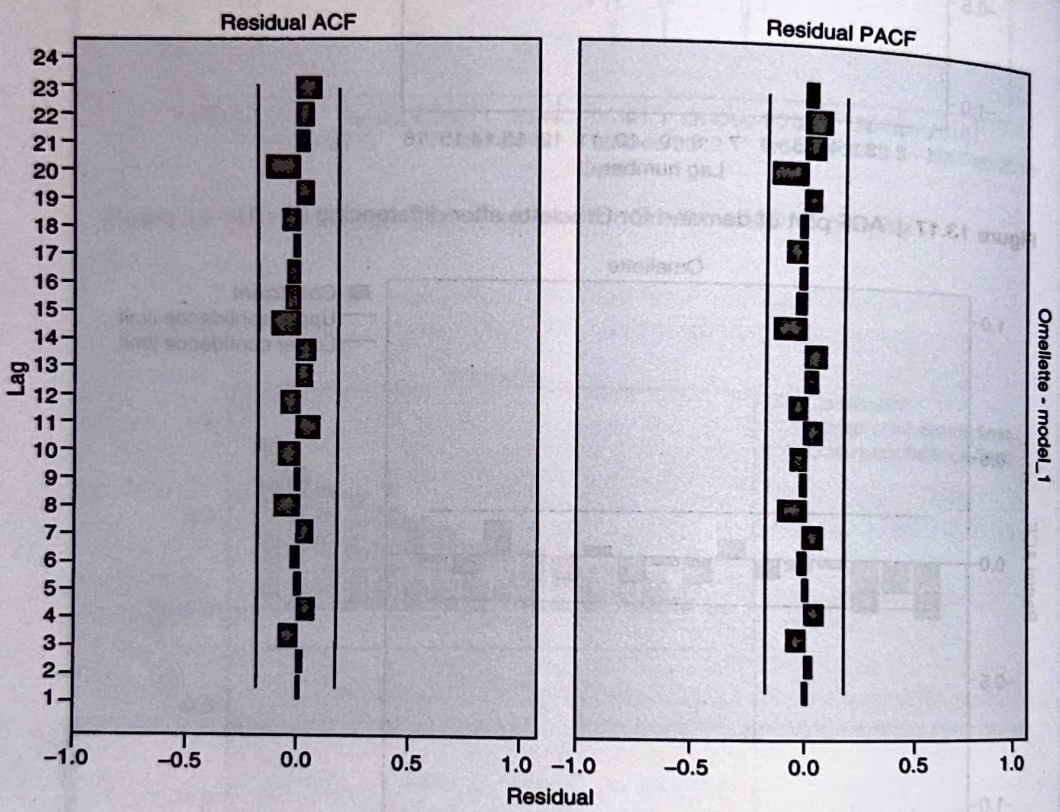


Figure 13.19 | ACF and PACF of residuals.

13.14.5 Ljung–Box Test for Auto-Correlations

Ljung–Box is a test of lack of fit of the forecasting model and checks whether the auto-correlations for the errors are different from zero. The null and alternative hypotheses are given by

H_0: The model does not show lack of fit

H_1: The model exhibits lack of fit

The Ljung–Box statistic (Q-Statistic) is given by (Ljung and Box, 1978)

$$Q(m) = n(n+2)\sum_{k=1}^{m}\frac{\rho_k^2}{n-k} \tag{13.53}$$

where n is the number of observations in the time series, k is the number of lag, ρ_k is the auto-correlation of lag k, and m is the total number of lags. Q-statistic is an approximate chi-square distribution with $m - p - q$ degrees of freedom where p and q are the AR and MA lags. The Q-statistic for ARIMA(1, 1, 1) is 10.216 (Table 13.29) and the corresponding p-value is 0.855 and thus we fail to reject the null hypothesis. $Q(m)$ measures accumulated auto-correlation up to lag m.

13.15 | Power of Forecasting Model: Theil's Coefficient

The power of forecasting model is a comparison between Naïve forecasting model and the model developed. In the Naïve forecasting model, the forecasted value for the next period is the same as the last period's actual value ($F_{t+1} = Y_t$). Theil's coefficient (U-statistic) is given by (Theil, 1965)

$$U = \frac{\sum_{t=1}^{n}(Y_{t+1} - F_{t+1})^2}{\sum_{t=1}^{n}(Y_{t+1} - Y_t)^2} \tag{13.54}$$

Theil's coefficient is the ratio of the mean squared error of the forecasting model to the MSE of the Naïve model. The value of $U < 1$ indicates that forecasting model is better than the Naïve forecasting model. $U > 1$ indicates that the forecasting model is not better than Naïve model. For the data shown in Table 13.14 (demand for avionic system spares), the U-statistic calculations are shown in Table 13.31.

Table 13.31 | U-statistic calculation

Day	Y_t	ARMA (1,2) Forecast	$(Y_t - F_t)^2$	Naïve Forecast ($F_{t+1} = Y_t$)	$(Y_t - F_t)^2$
31	503	464.8107	1458.423	443	3600
32	688	378.5341	95769.15	503	34225
33	602	444.6372	24763.04	688	7396
34	629	685.8851	3235.909	602	729
35	823	743.5124	6318.281	629	37636
36	671	630.7183	1622.614	823	23104
37	487	649.3491	26357.22	671	33856
		Total	159524.6	Total	140546

The U-statistic value = 159524.6 / 140546 = 1.1350. That is, ARMA(1, 2) model is not better than Naïve forecasting.

Summary

1. Forecasting is one of the important tasks carried out using analytics by many organizations since accurate forecasting is important for taking several decisions such as man-power planning, materials requirement planning, budgeting, and supply chain related issues.
2. Forecasting is carried out on a time-series data in which the dependent variable Y_t is observed at different time periods t.
3. Several techniques such as moving average, exponential smoothing, and auto-regressive models are used for forecasting the future value of Y_t.
4. The forecasting models are validated using accuracy measures such as RMSE, MAPE, AIC, and BIC.
5. Simple techniques such as moving average and exponential smoothing may outperform complex regression based

models in certain scenarios. Thus, it is important to develop forecasting models using several techniques before selecting the final model.

6. Regression model in the presence of independent variables may outperform other techniques.
7. Auto Regressive (AR) models are regression based models in which the dependent variable is Y_t and the independent variables are Y_{t-1}, Y_{t-2}, etc.
8. AR models can be used only when the data is stationary.
9. Moving Average (MA) models are regression models in which the independent variables are past error values.
10. Auto-Regressive Integrated Moving Average (ARIMA) has three components: Auto-regressive component with p lags – AR(p), moving average component with q lags – MA(q), and integration which is differencing the original data to make it stationary.
11. One of the necessary conditions of acceptance of ARIMA model is that the residuals should follow white noise.
12. In ARIMA, model identification, that is identifying the value of p in AR and q in MA, is achieved through Auto-Correlation Function (ACF) and Partial Auto-Correlation Function (PACF).
13. The stationarity of time-series data is usually checked using Dickey–Fuller and Augmented Dickey Fuller test.
14. The overall model accuracy of forecasting model is tested using Ljung–Box test.

Multiple Choice Questions

1. Seasonality in time-series data is caused due to
   (a) Changes in macro-economic factors such as recession, unemployment, and so on
   (b) Festivals and customs in a society
   (c) Random events that occur over a period of time
   (d) Changes in customer behaviour driven by new products and promotions
2. In a simple exponential smoothing method, the low value of smoothing constant α is chosen when
   (a) The data has high fluctuations around the trend line
   (b) There is seasonality in the data
   (c) The data is smooth with low fluctuations
   (d) There are variations in the data due to cyclical component
3. White noise is
   (a) Uncorrelated errors with expected value 0.
   (b) Uncorrelated errors that are constant and do not change with time.
   (c) Uncorrelated errors that follow normal distribution with mean 0 and constant standard deviation
   (d) Errors that follow normal distribution with constant mean and standard deviation
4. A stationary process in a time series is a process for which
   (a) Mean and variance are constant at different time points
   (b) The time series follows normal distribution with zero mean and constant standard deviation
   (c) The covariance of the time series depends only on the lag
   (d) Mean and standard deviation are constant at different time points and the covariance depends only on the lag between the values and is constant for a given lag
5. In a pure Auto-Regressive process, AR(p), the value of p can be identified using
   (a) Auto-Correlation Function
   (b) Partial Auto-Correlation Function
   (c) Auto-Correlation and Partial Auto-Correlation Function
   (d) Ljung–Box test
6. Power of a forecasting model is calculated using
   (a) Root Mean Square Error (RMSE)
   (b) Theil's coefficient
   (c) Mean Absolute Percentage Error (MAPE)
   (d) Bayesian Information Criteria (BIC)
7. A necessary condition for accepting a time-series forecasting model is
   (a) The residuals should follow a normal distribution
   (b) The residuals should be white noise
   (c) The residuals should be black noise
   (d) The residuals should follow a normal distribution and the R-square should be high
8. In an ARIMA model, differencing is carried out
   (a) To convert a stationary process to a non-stationary process
   (b) To convert a non-stationary process to a stationary process
   (c) To remove seasonal fluctuations from the data
   (d) To remove cyclical fluctuations from the data
9. Overall fitness of a forecasting model is checked using
   (a) Durbin–Watson Test (b) Theil coefficient
   (c) Ljung–Box test (d) Dickey–Fuller test
10. Presence of non-stationarity is checked using
    (a) Durbin–Watson Test (b) Theil coefficient
    (c) Ljung–Box test (d) Dickey–Fuller test

Exercises

1. Quarterly demand for certain parts manufactured by Jack and Jill company is shown in Table 13.32.

Table 13.32 | Quarterly demand

Year	Quarter	Value
2012	Q1	75
	Q2	60
	Q3	54
	Q4	59
2013	Q1	86
	Q2	65
	Q3	63
	Q4	80
2014	Q1	90
	Q2	72
	Q3	66
	Q4	85
2015	Q1	100
	Q2	78
	Q3	72
	Q4	93

(a) Calculate the seasonality index for different quarters using the first three years of data.

(b) Develop forecasting models using moving average, single exponential smoothing, and an appropriate ARMA model after de-seasonalizing the data (assume multiplicative model, $Y_t = T_t \times S_t$).

(c) Forecast the demand for 2015 (all four quarters) using moving average, exponential smoothing, and ARMA. Calculate RMSE, MAPE, and Theil's coefficient.

2. Data on monthly demand for a product over three years (between 2013 and 2015) is given in Table 13.33.

(a) Calculate the seasonality index using methods of averages.

Table 13.33 | Monthly demand

Month	2013	2014	2015
January	15	23	25
February	16	22	25
March	18	28	35
April	18	27	36
May	23	31	36
June	23	28	30
July	20	22	30
August	28	28	34
September	29	32	38
October	33	37	47
November	33	34	41
December	38	44	53

(b) De-seasonalize the data assuming that Y_t is product of trend and seasonality.

(c) Develop the best forecasting model by comparing MAPE of MA, ES, and ARMA models. Compare the models using MAPE and Theil's coefficient.

3. Television rating points of a program with over 30 episodes is shown in Table 13.34.

(a) Develop a forecasting model using regression $Y_t = \beta_0 + \beta_1 t$, where Y_t is the TRP at time t. Is there any trend in the data? Use the regression model developed to answer.

(b) Is there an auto-correlation in the data? Conduct an appropriate hypothesis test to justify your answer.

(c) The television channel would like to replace the program with a new program. The average TRP of new program will be eight points. Based on the model developed, comment whether they should replace the existing program with a new one.

(d) Calculate the probability that the TRP for episode 31 will be more than 10.

Table 13.34 | Television rating points

Episode	1	2	3	4	5	6	7	8	9	10
TRP	7.98	9.8	9.53	7.23	7.34	9.62	9.8	7.9	8.26	8.17
Episode	11	12	13	14	15	16	17	18	19	20
TRP	8.36	8.5	9.03	9.82	9.77	10.77	9.46	9.31	10.32	9.03
Episode	21	22	23	24	25	26	27	28	29	30
TRP	10.22	10.28	11.99	11.21	9.81	9.35	9.93	11.22	10.4	10.94

4. Auto-correlation function and partial auto-correlation functions for a dataset are shown in Figures 13.20 and 13.21, respectively.

(a) Based on ACF and PACF plot, what values of p and q are suitable for auto-regressive and moving average processes?

(b) The ACF and PACF of residuals are shown in Figure 13.22. What can you conclude from Figure 13.22 about the model?

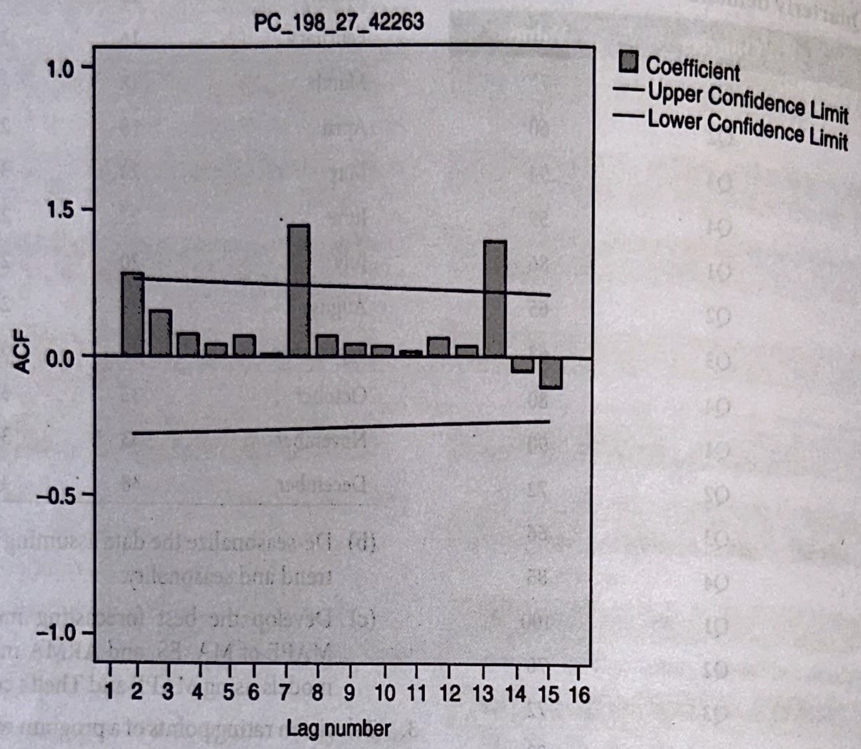


Figure13.20 | ACF Plot.

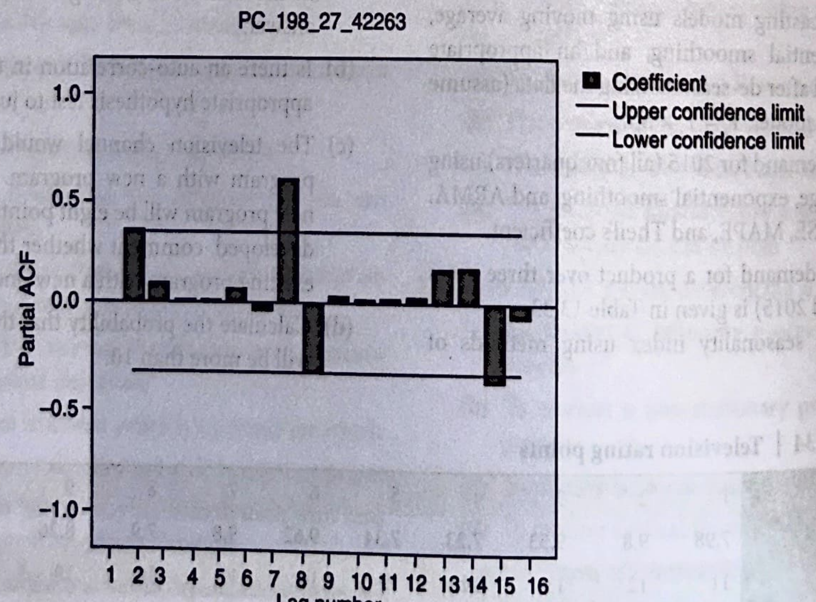


Figure 13.21 | PACF plot.

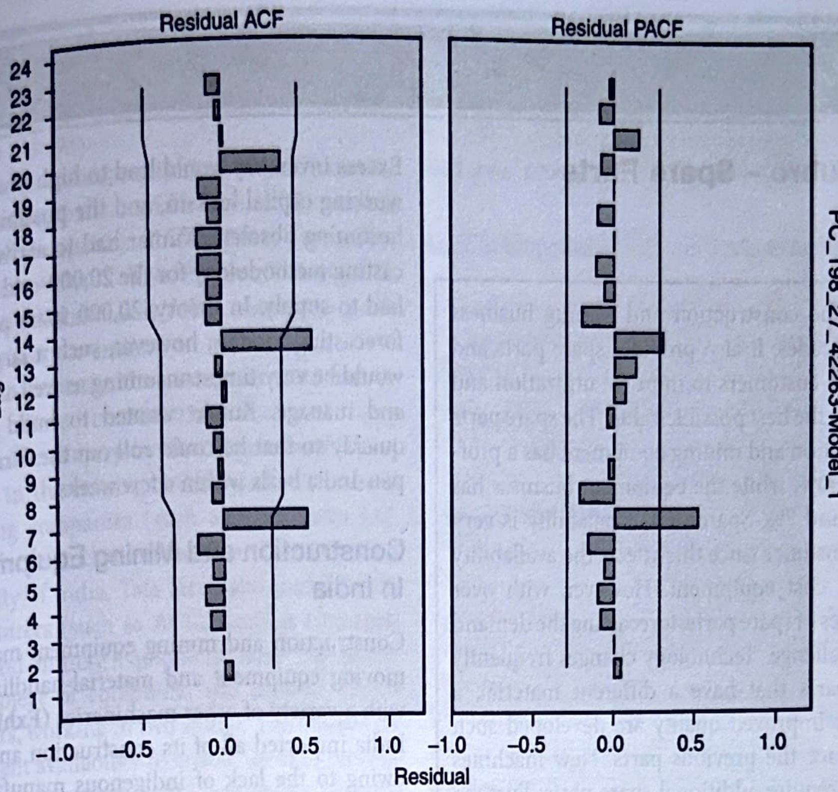


Figure 13.22 | ACF and PACF of residuals.

5. Table 13.35 shows the volume of sales in a retail store on a day and snow fall in inches in the region.
   (a) Develop a forecasting model using moving average, single exponential smoothing, regression and AR(2). Calculate the MAPE for all four models. Which model gives the least MAPE?
   (b) Construct ACF and PACF plot. Develop an ARIMA model [if the model is different from AR(2) model developed in (a)].
   (c) Which model would you recommend to the retail store for forecasting volume of sales?

Table 13.35

Day	1	2	3	4	5	6	7	8	9	10
Sales	72462	607500	816150	973300	744180	255665	1014410	464105	848775	1182225
Snow fall	16.2	20.9	24.84	18.52	5.83	26.74	10.95	22.05	30.35	15.63
Day	11	12	13	14	15	16	17	18	19	20
Sales	646825	656760	597360	559205	159470	225250	1105890	879610	718800	1000875
Snow fall	16.68	15.04	14.11	3.5	5.82	29.18	21.9	18	25.73	18.83
Day	21	22	23	24	25	26	27	28	29	30
Sales	757585	558160	899260	975355	1195165	542225	687160	220905	988275	931930
Snow fall	13.88	23.24	24.77	30.55	12.83	17.64	4.95	26.09	23.46	4.57
Day	31	32	33	34	35	36	37	38	39	40
Sales	218295	387140	289410	448985	1031180	147510	649140	316935	861060	287745
Snow fall	10.08	7.18	11.59	26.88	0.57	17.28	7.53	22.56	6.47	12.77

Case Study

Larsen and Toubro – Spare Parts Forecasting[3]

> L&T has been in the construction and mining business for the last three decades. It also provides spare parts and after-sales service to customers to improve utilization and helps them to obtain the best possible value. The spare parts business of construction and mining equipment has a profitability of around 30%, while the equipment business has profitability of around 7%. Spare parts availability is very important to the consumer since this affects the availability of the high-capital cost equipment. However, with over 20,000 different types of spare parts, forecasting the demand of each part is a challenge. Technology changes frequently and superseding parts that have a different material, a different design, or improved quality are developed such that these can replace the previous parts. New machines that are developed require additional spare parts. Further, demand is quite seasonal since most of the customers plan their annual overhaul during the monsoon season, when construction is virtually at a standstill.
>
> – *Vijaya Kumar, DGM, L&T Construction and Mining Business (April 2014)*

Monday, April 21, 2014: Vijaya Kumar walked out of his cabin trying to clear his head that was filled with numbers. He was going over the demand trends and forecasts for around 20,000 spare parts of the various construction and mining equipment sold by Larsen and Toubro (L&T). Vijaya Kumar was the Deputy General Manager of the Supply Chain Department of L&T's Construction and Mining Business (CMB). L&T has been India's largest technology, engineering, construction, and manufacturing company. CMB provided heavy construction and mining equipment to its customers, along with support services and spare parts. The supply of spare parts was critical since the customer would face severe losses in the event of equipment unavailability. Forecasting was done *ad hoc*, based on the experience of the planning personnel. The value of each spare part ranged from ₹10 to ₹8 million (USD 1 ≅ ₹60 in April 2014). Maintaining the balance of the spare parts inventories was critical since unavailability would result in loss of revenues, decreased profitability, and increased customer dissatisfaction, and would also give rise to the spurious products industry. Excess inventory would lead to high inventory carrying costs, working capital lock-in, and the possibility of the spare parts becoming obsolete. Kumar had to arrive at an accurate forecasting methodology for the 20,000-odd spare parts that CMB had to supply. In theory, 20,000 spare parts called for 20,000 forecasting models; however, such a large number of models would be very time-consuming as well as expensive to develop and manage. Kumar wanted to build a forecasting model quickly so that he could roll out the forecasting strategy on a pan-India basis within a few weeks.

Construction and Mining Equipment Industry In India

Construction and mining equipment mainly included earth-moving equipment and material-handling equipment, along with a variety of other machineries (**Exhibit 1**). Prior to 1960, India imported all of its construction and mining equipment owing to the lack of indigenous manufacturing facilities. In 1964, Bharat Earth Movers Ltd. (BEML), a public sector enterprise, was established to produce dozers and dumpers under technology license from LeTorneu Westinghouse (USA) and Komatsu (Japan). Another private sector enterprise, Hindustan Motors, forayed into this sector in 1969. Hindustan Motors had technological collaboration with Terex, UK. Several foreign players such as Case, Caterpillar, Ingersoll Rand, Komatsu, JCB, Hitachi, Volvo, and Lieber entered the country after the 1991 economic reforms, either through joint ventures with Indian companies or by setting up wholly owned subsidiaries.

The industry size was estimated to be around ₹153 billon (USD 2.55 billion); it had grown at a CAGR of around 10% during the fiscal period of 2009–2013 owing to a high growth of 33% during FY 2009–2010. The annual growth rate was around 3.4% during 2010–2013.[4] The growth of the construction and mining equipment industry depended on construction and mining activities in the country, which had considerably slowed down owing to economic slowdown, high interest rates, and the slowdown in policy making by the Indian government. The construction and mining sectors reported growth of around 5.9% and 0.4%, respectively, in FY 2012–2013. The construction sector comprised residential and commercial real estate along with infrastructure construction. The demand for residential and commercial estate had decreased. There was a slowdown in the infrastructure sector owing to delays in obtaining government clearances, trouble with land acquisition, and the high cost of capital. The mining sector largely included coal, iron ore, and limestone mining; this sector was affected by regulatory uncertainties. The proposed infrastructure spending of USD 1 trillion under the 12th Five Year Plan (2012–2017) and regulatory clarity over mining licenses were expected to revive

[3] Copyright © Indian Institute of Management, Bangalore. The case was authored by Suhruta Kulkarni, Prakash Hegde Ruchi Jaiswal and U Dinesh Kumar, Professor of Quantitative Methods and Information systems prepared this case for classroom discussion. This case is not intended to serve as an endorsement or source of primary data, or to show effective or inefficient handling of decision or business processes. The case was published at the Harvard Business Publishing as part of the IIMB's case collection in 2015. Reproduced with the permission of IIM Bangalore.

[4] Source: India's Construction and Mining Equipment Industry, D&B, August 2013.

the construction and mining industry; the projected growth in the infrastructure sector was estimated to be not more than 5% per annum. Similar growth was expected in the construction and mining equipment industry.

The construction and mining equipment industry mainly involved large players owing to the need for high upfront capital investments and technical expertise. BEML, L&T, Caterpillar, JCB India, Komatsu, Case, Volvo, Telcon, Escorts, Tata-Hitachi, Kobelco, Liebherr, Ingersoll Rand, and Voltas were some of the major players in the industry in India.

In the construction industry, the customers were mainly unorganized players, since construction activities were sub-contracted to smaller players by the large construction companies. However, in the mining industry, the customers were large coal mining companies (such as Coal India Ltd. and its subsidiaries), large steel manufacturers (for iron ore) (such as Steel Authority of India, Tata Steel, and Jindal Steel), and cement manufacturers (such as ACC, Ambuja Cements, etc.). Construction and mining equipment were expensive, and the life of each piece of equipment was around 20,000 hours (around 3.5 years, working in two shifts). The customers required high equipment availability to enable them to recover their capital costs. The maintenance and availability of spare parts were critical for ensuring equipment availability. Further, genuine spare parts from the Original Equipment Manufacturer (OEM) had to be used to avoid damage to the equipment. Some customers purchased spare parts from local vendors for one of two reasons – unavailability of spare parts with the OEM, or the lower cost of the local/spurious spare parts, which affected the performance of the equipment. Thus, service and spare parts were critical elements of the construction and mining equipment industry.

L&T: The Indian Engineering Giant

L&T was founded in Bombay (Mumbai) in 1938 by two Danish engineers, Henning Holck-Larsen and Soren Kristian Toubro. They started their operations by representing Danish dairy equipment manufacturers. World War II stopped Danish supplies, which forced the founders to manufacture equipment indigenously. The war created opportunities for repairing ships, which led to the formation of repair and fabrication shops. During 1944–1946, L&T entered into several foreign collaborations. After India gained independence in 1947, there was a growing demand for equipment across industries. L&T expanded and set up offices across the eastern (Calcutta), southern (Madras), and northern (New Delhi) regions of India. In 1950, L&T became a public company, and it grew rapidly in the 1960s. Toubro and Larsen retired in 1962 and 1978, respectively. The company grew into an engineering major under the guidance of several eminent leaders.

In 2014, L&T established its presence in several businesses – infrastructure, defense, aerospace, hydrocarbons, heavy engineering, construction, power, mining and metallurgy, electrical and automation products, machinery and industrial products, information technology, financial services, ship building, and railway projects. L&T had design-to-build capacities for most of its businesses. It had 137 subsidiaries and 16 associated companies; it had not only a pan-India presence but also a global presence across 30 countries. The L&T Group had gross revenues of ₹752 billion (USD 13 billion) in FY 2013.[5]

L&T's Construction and Mining Business

The Construction and Mining Business (CMB) formed a part of the Machinery and Industrial Products (MIP) business at L&T (**Exhibit 2**). The MIP segment earned revenues of ₹23 billion, which formed around 4% of L&T's total revenues. However, the MIP segment had gross profit margins of around 16.3% as against L&T's overall gross profit margin of 13.2%.[6] Thus, even though MIP was a small segment in terms of revenues, it was important from the profitability perspective. CMB was formed in 1998 as a 50:50 joint venture between L&T and Komatsu Asia & Pacific, Singapore, which was a wholly owned subsidiary of Komatsu, Japan. In April 2013, L&T bought out Komatsu's 50% stake.[7]

L&T's CMB sold equipment such as dozer shovels, dozers, dumpers, hydraulic excavators, motor graders, pipe layers, surface miners, tipper trucks, wheel dozers, and wheel loaders (**Exhibit 3**). CMB also provided equipment installation and commissioning services as well as other maintenance services. CMB did not manufacture the equipment that it sold and serviced; it had tie-ups with OEMs such as Komatsu, Scania, and so on for sourcing the equipment. Additionally, CMB sold and serviced construction and mining equipment that was manufactured in other L&T divisions.

CMB involved different verticals such as sales and marketing, services, and supply chain. The sales and marketing vertical was responsible for marketing and completing the sale orders that were placed with the OEMs. The OEM invoiced the customer and dispatched the equipment. Subsequently, CMB's service team would install and commission the equipment for the customer. The service team also took care of servicing during the warranty period and dealt with any other kind of servicing-related assistance that the customer required. The supply chain team was responsible for the availability of spare parts and the related internal logistics.

Spare Parts Supply Chain at CMB

CMB supplied parts across the country through its central warehouse in Nagpur. The CMB group had four fully equipped service stations in Chennai, Delhi, Durgapur, and Pune, which catered to the southern, northern, eastern, and western regions of the country, respectively. The facility in Nagpur catered to the central region. There were 58 sub-service centres under the fully equipped service centres. In addition to this, CMB had

5 Source: http://www. larsentoubro. com/.

6 Source: L&T Annual Report, 2012–13.

7 Source: http://articles. economictimes. indiatimes. com/2013-04-13/news/38511372_1_larsen-toubro-construction-equipment-brand-equity

26 dealers who had 84 outlets across the country. Spare parts of the construction equipment were generally supplied through the dealers, while mining equipment spare parts were supplied via a mix of service centres and dealer outlets.

Spare parts were classified into six main categories: filters, engine maintenance parts, seals and hoses, v-belts, undercarriage, and ground engaging tools (**Exhibit 4**). Each category included different types of spare parts, and each type consisted of several varieties with different technical specifications. On average, a major piece of equipment would consist of around 1,100 unique parts. Thus, there were more than 20,000 spare parts across the different types of equipment sold by L&T.

The demand for the spare parts would vary owing to a variety of reasons. Some spare parts would be required because of operational wear and tear, while others would be needed owing to breakdown. Customers followed different maintenance strategies such as preventive or breakdown maintenance, which caused a variation in demand. CMB provided warranties for the equipment; in the instance of any breakdown during the warranty period, spare parts would be required. Warranties were generally provided for 3,000 hours or 1 year of equipment operations; the typical life of each piece of equipment was around 20,000 hours. Further, several customers opted for extended warranty, which added to the variation in demand. During the extended warranty period, L&T had to support the machine similar to what was done during the warranty period. Annual Maintenance Contracts (AMC) guaranteed the availability of equipment to customers. Availability of spare parts played a critical role in fulfilling AMC services. Additionally, the sales and marketing team sometimes offered spare parts free of cost. The sales team would offer some free spare parts to the customers as per the prevailing offers made by the competition. Some of the customers would not opt for these offers and would demand equivalent value. In such instances, L&T would offer them equivalent value for the future purchase of any other spare parts from L&T (for this value). This was similar to providing coupons that were equal in value to the free spare parts offered by L&T. These coupons could be redeemed for spare parts purchased from L&T. The customer benefitted from the option of using the value gained from the 'free of cost' offer for future purchases. However, the purchases made by the customers using such coupons were generally one-time purchases, and the probability of the repetition of this demand was low. Thus, the variability in demand increased with such 'free of cost' offers.

After-sales business in the machine and plant construction industry generally accounted for approximately 25% of the total sales (with two-thirds being derived from the sales of spare parts and one-third from services), while it accounted for almost 50% of the total profits.[8] At CMB, spare parts contributed 25% of the total revenues; the rest was contributed by equipment sales. The profitability of the spare parts business was around 30%, while the profitability of the equipment sales business was around 7%. The unavailability of spare parts led to not only the loss of a transaction but also the potential loss of a customer, since the customer preferred immediate replacement of the damaged spare parts to avoid losses to his/her business.

Forecasting Demand for Spare Parts

Kumar was looking at the data for the period April 2009–April 2013. He had monthly details of the demand for each of the spare parts. It was difficult and very expensive to forecast the individual demand for 20,000 spare parts. Therefore, the CMB team had categorized the spare parts in several ways. The first categorization was based on the frequency of demand, according to which three categories – fast (F), medium (M), and slow (S) – were formed. This was known as the FMS categorization. The second categorization was based on the value of the product. Three categories were formed according to value: high (H), medium (M), and low (L); this was known as the HML categorization. The third categorization was based on the well-known ABC analysis according to which the spare parts belonging to category A constituted 70% of the sales, those belonging to category B constituted 25% of the sales, while category C spare parts constituted 5% of the sales. The spare parts were further classified based on combinations of these three categories and were ranked accordingly. For example, a category A spare part that was fast-moving (F) and had a high value (H) was assigned the top rank; this combination was termed 'AHF' (**Exhibit 5**). According to Kumar:

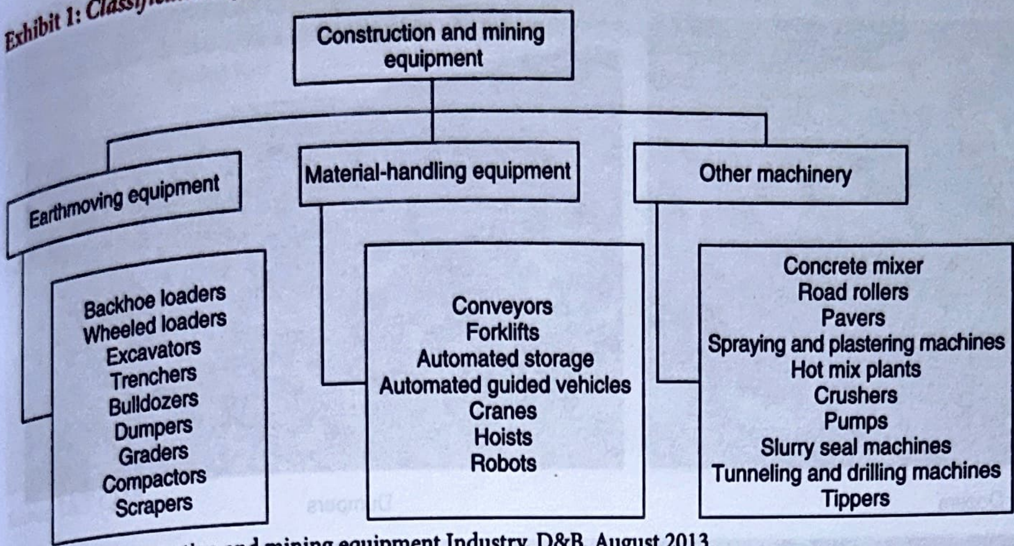
> We used several analytical tools such as exponential smoothing, autoregressive integrated moving average (ARIMA) model, and Croston's method for forecasting the different categories of spare parts. We figured that one forecasting model would not be suitable for all the spare parts since the pattern of demand varied across the spare parts.

Kumar's team wanted to develop forecasting models for spare parts that would result in less than 10% error. However, Kumar wondered whether it was possible to develop such a model for all the spare parts and whether this model would work in the face of constantly changing industry trends. Was there any additional data that could be incorporated into the model to make it more robust? What strategies should be followed by the CMB team to ensure better availability and reduced inventory costs?

---

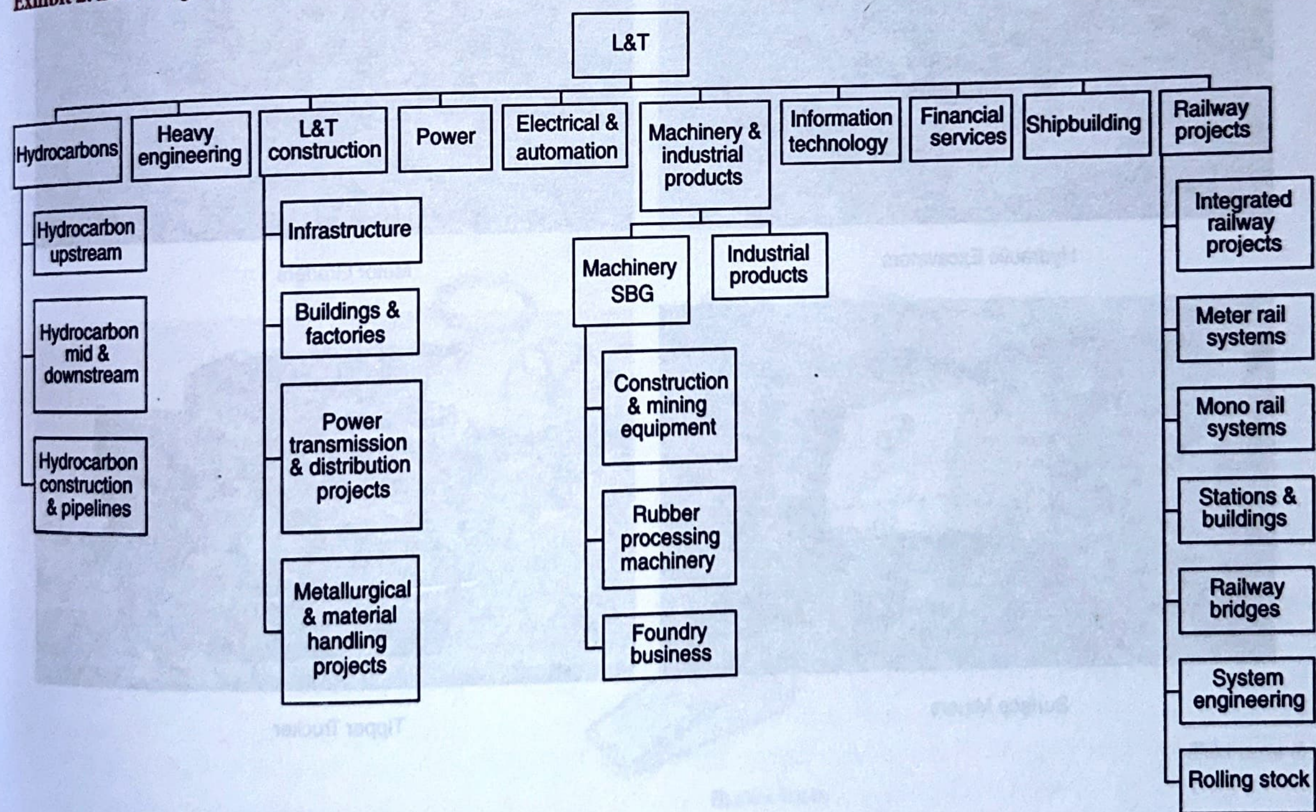
[8] Stephen M. Wagner, Ruben Jonke, and Andreas B. Eisingerich, A Strategic Framework for Spare Parts Logistics, *California Management Review*, 2012, 54(4), 69.

Exhibit 1: Classification of construction and mining equipment.



Source: India's construction and mining equipment Industry, D&B, August 2013.

Exhibit 2: L&T's organizational structure.

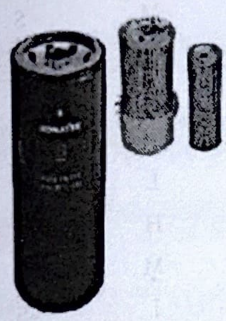


Source: L&T.

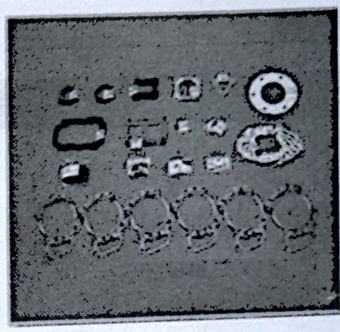
Exhibit 4: L&T CMB's spare parts

Filters	Engine Maintenance Parts	Seals & Hoses	Undercarriage	Ground Engaging Tools	V-Belts
Engine Filters	Gasket Kits	Hydraulic Hoses	Track-Links	Bucket Teeth	V-Belts
1000-Hour Filters	Rain Caps	O-Rings	Sprocket Teeth	Cutting Edges	
Fuel Filters	Exhaust Pipe	Oil Seals	Track Shoes	Blades	
Oil Filters	Mufflers	Fuel Hoses	Rollers	Side Cutters	
Air Cleaner	Cylinder Liners	Dust Seals	Idlers		
Corrosion Resistor	Pistons	Seal Washers			
Hydraulic Filter	Piston Rings	Low Pressure Heads			
	Thermostat	Water Hoses			
	Fuel Water Separators	Hose Clamps			
		Back-up Rings			

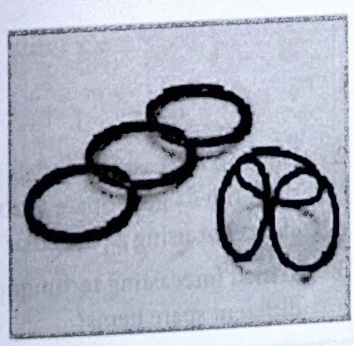
Source: L&T CMB.



Fuel Filters

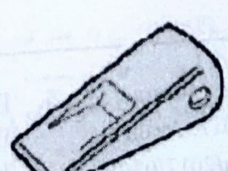


Gasket Kit



O-Rings

Rollers



Bucket Tooth

Exhibit 5: ***Categorization of spare parts***

FMS Categorization (Demand)	
Fast (F)	Demand > 8 per month
Medium (M)	8 per month ≥ Demand > 4 per month
Slow (S)	Demand ≤ 4 per month

ABC Categorization (Sales)	
A	70% of sales
B	25% of sales
C	15% of sales

HML Categorization (Value)	
High (H)	Value ≥ ₹50,000
Medium (M)	₹50,000 > Value > ₹ 10,000
Low (L)	Value ≤ ₹10,000

Exhibit 5: ***(Continued)***

ABC	HML	FMS	Comb	Rank
A	H	F	AHF	1
A	M	F	AMF	2
A	L	F	ALF	3
B	H	F	BHF	4
B	M	F	BMF	5
B	L	F	BLF	6
C	H	F	CHF	7
C	M	F	CMF	8
C	L	F	CLF	9
A	L	M	ALM	10
B	M	M	BMM	11
B	L	M	BLM	12
C	M	M	CMM	13
C	L	M	CLM	14
A	H	M	AHM	15
A	M	M	AMM	16
B	H	M	BHM	17
C	H	M	CHM	18
A	H	S	AHS	19
A	M	S	AMS	20
A	L	S	ALS	21
B	H	S	BHS	22
B	M	S	BMS	23
B	L	S	BLS	24
C	H	S	CHS	25
C	M	S	CMS	26
C	L	S	CLS	27

Source: L&T construction and mining business

Case Questions (Use the Data provided)

1. What strategy should Vijaya Kumar adopt for developing forecasting model for demand estimation of 20,000 spare parts?
2. Develop forecasting models for data provided in the Excel sheet titled "L&T Spare Parts Forecasting" and discuss the choice for using a particular forecasting model.
3. Which forecasting techniques should L&T use to forecast different spare items?

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